

Project Design Phase-II

Data Flow Diagram & User Stories

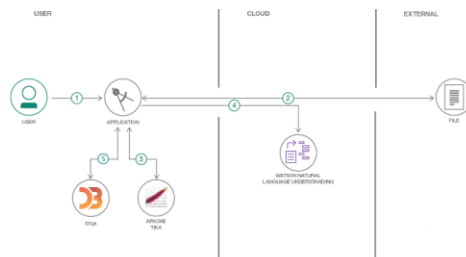
Date	18 June 2026
Team ID	
Project Name	ShopEZ
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: [\(Simplified\)](#)

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

User Stories

User Stories User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Visitor	Authentication	USN-1	As a visitor, I want to register an account, so that I can save items in my cart.	Password is hashed securely with bcryptjs.Email must be completely unique in the database.Client securely stores the session JWT locally.	High	Sprint-1
User / Admin	Authentication	USN-2	As a user, I want to be routed dynamically based on role, so that I don't see admin metrics if I am a customer.	Token verification occurs seamlessly on entry.Redirects Admins to /admin and Customers to /.Protects all administrative routes with a 403 authorization filter.	High	Sprint-1
Shopper	Customer Experience	USN-3	As a shopper, I want to query catalog items by text and filter fields, so that I can locate products quickly.	Text matching successfully searches titles and descriptions.Sidebar filters apply Category and Gender choices cleanly.Sorting updates product listings instantly by price or discount margins.	High	Sprint-1
	Customer Experience	USN-4	As a shopper, I want to select sizes and toggle carousel image views, so that I can inspect products.	Clicking carousel thumbnails successfully swaps the main image display.Sizes (S, M, L, XL) toggle active state layout indicators.Quantity modifier locks below 1 and disables at max stock.	Medium	Sprint-1

Customer	Customer Experience	USN-5	As a customer, I want to view itemized invoices in the cart, so that I know exactly what I am saving.	Subtotal automatically recalculates on quantity adjustments.Discounts subtract mathematically correctly.Cart explicitly lists free shipping threshold boundaries.	High	Sprint-1
	Fulfillment	USN-6	As a checkout customer, I want to complete checkout details, so that my order is logged for COD delivery.	Validates phone numbers (10 digits) and pincodes accurately.Creates distinct orders records per product unit in the cart.Clears active cart contents automatically on success.	High	Sprint-1
	Fulfillment	USN-7	As a customer, I want to cancel active orders on my profile, so that I can stop unwanted deliveries.	Cancellation button is only visible for orders in an "order placed" state.Updates status immediately in the database to "cancelled".Customers are blocked from cancelling other users' orders.	Medium	Sprint-2
Administrator	Store Management	USN-8	As an admin, I want to manage product stock listings (CRUD), so that catalog items remain updated.	Admin panel provides explicit edit and delete icons on table listings.Product form strictly validates input fields (price > 0, valid main image URL).Submission triggers an instant update on the CatalogPage view.	High	Sprint-2

Data Flow Diagrams:(Level 0)



Level 1:

